

Notes: Pierce and Schott (AER, 2016)

The Surprisingly Swift Decline of US Manufacturing Employment

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1 Big picture

- **Question.** Why did US manufacturing employment fall so sharply after 2000? More specifically, did the United States' grant of Permanent Normal Trade Relations (PNTR) to China contribute to the decline?
- **Core puzzle.** PNTR did *not* lower the tariff rates actually applied to Chinese imports. China had already been receiving the relatively low Normal Trade Relations (NTR) rates through annual renewal. What changed was that PNTR eliminated the possibility that tariffs would jump back up to the much higher non-NTR rates.
- **Main idea.** Even if current tariff rates are unchanged, eliminating the risk of future tariff spikes can matter a lot when firms must pay sunk costs to build sourcing relationships, move production offshore, expand exporting, or reorganize production.
- **Empirical strategy.** The paper exploits cross-industry variation in the *NTR gap* — the difference between the high non-NTR tariff and the low NTR tariff — in a generalized difference-in-differences design. Industries with larger gaps were more exposed to the policy change.
- **Main result.** Industries with larger NTR gaps experience larger employment declines after PNTR. In the baseline specification, the implied relative decline in manufacturing employment is about **0.15 log points** on average.
- **Trade response.** High-gap products see larger increases in imports from China, more US importers, more Chinese exporters, and more importer-exporter pairs. Related-party trade also rises, which is consistent with offshoring inside firms.
- **Production response.** The policy is associated with shifts toward less labor-intensive production: higher industry skill and capital intensity, relative capital deepening within continuing plants, and lower employment and higher exit in plants exposed directly or through downstream customers.
- **Why this paper matters.** The paper shows that **trade policy uncertainty** can have first-order real effects. It also reframes the early-2000s manufacturing collapse as partly a response to the resolution of uncertainty about access to China rather than only a response to contemporaneous tariff cuts.

2 Data and measurement (key objects)

2.1 US manufacturing employment and plant data

- **Main dataset.** The core employment data come from the US Census Bureau's **Longitudinal Business Database (LBD)**, which tracks virtually every establishment with employment in the private nonfarm economy annually as of March 12.
- **Unit of observation.** Depending on the specification, the paper works at the **industry-year** level or the **plant-year** level.
- **Sample period.** The main employment sample spans **1990 to 2007**.
- **Constant manufacturing sample.** Because industry codes shift from SIC to NAICS over time, the paper uses time-consistent industry families so that changes in employment are not mechanical artifacts of reclassification.

- **Outcome variables.** The main outcome is log employment. Additional outcomes include production-worker employment, non-production-worker employment, hours, capital, plant death, and measures of factor intensity.

2.2 Policy exposure: the NTR gap

- **Definition.** Exposure to PNTR is measured using the **NTR gap**

$$\text{NTR Gap}_i = \text{Non-NTR Rate}_i - \text{NTR Rate}_i.$$

- **Construction.** Tariff rates are observed at the eight-digit HS level and then aggregated to industries using BEA concordances.
- **Magnitude.** In 1999, the average NTR gap across industries is about **0.33**, with a standard deviation of about **0.14**. Average NTR rates are about **4 percent**, while average non-NTR rates are about **37 percent**.
- **Why it is useful.** The gap measures how much tariff uncertainty was resolved by PNTR. Industries with larger gaps had more to gain from the removal of downside tariff risk.

2.3 Controls and alternative explanations

- **Initial industry characteristics.** The paper controls for 1990 capital intensity, 1990 skill intensity, contract intensity, and whether the industry produces advanced-technology goods.
- **Time-varying controls.** These include MFA quota exposure, NTR tariff rates, and union membership.
- **China-side reforms.** The regressions also account for changes in Chinese import tariffs, Chinese production subsidies, and Chinese export licensing requirements associated with China's WTO accession.
- **Business-cycle concerns.** Additional robustness checks incorporate controls meant to soak up industry-specific sensitivity to aggregate demand and exchange-rate movements.

2.4 Trade, international comparison, and factor-intensity data

- **US import transactions.** The paper uses the Census Bureau's **Longitudinal Foreign Trade Transactions Database (LFTTD)** from **1992 to 2007**. These data identify the product, source country, US importer, foreign exporter, and whether the transaction is between related parties.
- **Chinese export data.** Chinese customs data provided via the National Bureau of Statistics track exports by **firm, product, destination, and year** from **2000 to 2005**, including ownership type and whether exports are general trade or processing-and-assembly trade.
- **European comparison sample.** The paper uses **UNIDO INDSTAT 4** employment data for the United States and European Union from **1998 to 2005** to compare the US experience against a place where policy did not change.

- **Factor-intensity data.** Industry and plant responses in capital and skill intensity are examined using the **Census of Manufactures** for 1992, 1997, 2002, and 2007.
- **Input-output linkages.** Upstream and downstream NTR gaps are constructed using BEA input-output tables.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism: uncertainty, sunk costs, and adjustment

- The paper is motivated by the standard **investment-under-uncertainty** logic. If firms must pay sunk costs to offshore, source from China, enter export relationships, or reorganize production, then a reduction in policy uncertainty raises the value of taking those actions.
- PNTR therefore can matter even without a change in current tariffs because it removes the possibility that annual NTR renewal fails and tariffs jump back to the much higher non-NTR schedule.
- This logic implies several margins of adjustment:
 - more US sourcing from China,
 - more Chinese entry into the US market,
 - more offshoring by US firms,
 - more capital- and skill-intensive production in the United States,
 - spillovers through supply chains.

3.2 The NTR-gap exposure measure

The paper’s key treatment intensity variable is

$$\text{NTR Gap}_i = \text{Non-NTR Rate}_i - \text{NTR Rate}_i.$$

- **Interpretation.** This is not a realized tariff change; it is the size of the *potential* tariff increase that PNTR ruled out.
- **Economic meaning.** A larger gap means more uncertainty was removed and larger expected gains from investing in China-related activity.
- **Identification relevance.** Industries with larger NTR gaps should respond more strongly if PNTR is the relevant policy shock.

3.3 Baseline industry-level difference-in-differences

The main estimating equation is

$$\ln(\text{Emp}_{it}) = \theta \text{PostPNTR}_t \times \text{NTR Gap}_i + \text{PostPNTR}_t \times X'_i \gamma + X'_{it} \lambda + \delta_t + \delta_i + \alpha + \varepsilon_{it}.$$

- **Dependent variable.** Log employment in industry i and year t .
- **Treatment term.** $\text{PostPNTR}_t \times \text{NTR Gap}_i$ is the DID term of interest.
- **Post period.** The baseline treats years from **2001 onward** as post-PNTR.
- **Fixed effects.** Industry fixed effects absorb permanent industry differences; year fixed effects absorb common aggregate shocks.
- **Weights.** Regressions are weighted by **1990 industry employment**.
- **Interpretation of θ .** A negative θ means industries more exposed to the removal of tariff uncertainty lose more employment after PNTR.

3.4 Event-study timing specification

To test for pre-trends, the paper replaces the single post indicator with year-by-year interactions:

$$\ln(\text{Emp}_{it}) = \sum_y \theta_y 1\{y = t\} \times \text{NTR Gap}_i + \sum_y 1\{y = t\} \times X'_i \beta_y + X'_{it} \lambda + \delta_t + \delta_i + \alpha + \varepsilon_{it}.$$

- **Goal.** Check whether high-gap industries were already on a different employment trend before PNTR.
- **Interpretation.** The identifying assumption becomes visually testable: before 2001, the θ_y should be near zero; after 2001, they should turn negative if PNTR matters.

3.5 United States versus European Union triple difference

To compare the United States to a control region where policy did not change, the paper estimates

$$\ln(\text{Emp}_{ict}) = \theta \text{PostPNTR}_t \times \text{NTR Gap}_i \times US_c + \delta_{ct} + \delta_{ci} + \delta_{it} + \alpha + \varepsilon_{ict}.$$

- **Third difference.** The United States is compared to the European Union.
- **Logic.** If the employment decline is really about the US change in China policy, then it should appear in the United States but not in the EU, which had already granted China permanent MFN-like treatment in 1980.

3.6 Trade-response specification for US imports

For trade margins, the paper estimates

$$O_{hct} = \theta 1\{c = \text{China}\} \times \text{PostPNTR}_t \times \text{NTR Gap}_h + \lambda \text{Tariff}_{hct} + \delta_{ct} + \delta_{ch} + \delta_{ht} + \alpha + \varepsilon_{hct},$$

where O_{hct} is one of several trade outcomes.

- **Outcomes.** Log import value, number of US importers, number of Chinese exporters, and number of importer-exporter pairs.
- **Interpretation.** A positive θ means that products more exposed to PNTR experienced faster growth of imports from China relative to other source countries after the policy change.

- **Related-party version.** The same design is used with related-party trade outcomes to study offshoring within firm boundaries.

3.7 Chinese export specification

The Chinese export regressions use the same triple-difference logic, but replace the China-source indicator with a US-destination indicator and exploit Chinese firm-level export data.

- **Purpose.** Determine whether the surge in trade is driven mainly by domestic Chinese firms or by foreign-owned firms operating in China.
- **Why it matters.** Stronger responses among foreign-owned firms are consistent with offshoring by multinational firms, potentially including US firms.

3.8 Factor-intensity regressions

The paper applies the same DID logic to industry and plant outcomes from the Census of Manufactures.

- **Industry outcomes.** Skill intensity, capital intensity, production employment, non-production employment, total employment, capital, and hours.
- **Plant outcomes.** The analogous measures are estimated for continuing plants, using plant-level NTR gaps computed as weighted averages of the gaps in plants' activities.
- **Interpretation.** These regressions distinguish between employment decline due to import substitution and employment decline due to domestic plants becoming more capital or skill intensive.

3.9 Input-output linkage specification

To study propagation through supply chains, the paper estimates

$$O_{pt} = \sum_{m \in \{\text{Own}, \text{Upstream}, \text{Downstream}\}} \theta_m \text{PostPNTR}_t \times \text{NTR Gap}_p^m + \text{controls} + \delta_t + \delta_p + \alpha + \varepsilon_{pt},$$

where O_{pt} is either log plant employment or a plant-death indicator.

- **Own gap.** Direct exposure through the plant's own industry.
- **Upstream gap.** Exposure through suppliers.
- **Downstream gap.** Exposure through customers.
- **Interpretation.** This design tests whether the policy shock transmits beyond directly exposed industries through production networks.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the NTR gap is plausibly exogenous

- Most of the variation in the NTR gap comes from **non-NTR tariff rates set in 1930** under Smoot-Hawley. The paper states that about **79 percent** of the cross-industry variation comes from these old non-NTR rates.
- That historical origin makes it hard to argue that the post-2000 employment decline caused the policy exposure measure.
- Moreover, if low-NTR rates had been raised in the past to protect declining industries, that would *reduce* the NTR gap, biasing estimates *against* finding an effect of PNTR.

4.2 Parallel trends and timing

- The event-study specification shows that coefficients relating the NTR gap to employment are close to zero before 2001 and become increasingly negative only after PNTR.
- This is exactly what the design requires: no differential trend before treatment, followed by a break after treatment.
- The timing therefore supports the interpretation that PNTR, rather than a longstanding trend, is driving the result.

4.3 Ruling out alternative explanations

- The paper controls for several China-side reforms that occurred around WTO accession: Chinese tariff cuts, licensing liberalization, subsidies, and foreign-investment-related contract intensity.
- It also controls for US-side forces that could be confounded with the timing: MFA quota removal, industry unionization, and exposure to the technology boom-bust episode.
- Additional robustness checks show similar results when adding explicit business-cycle controls, using revealed tariff measures, or using the NTR gap measured in 1990 instead of 1999.
- The authors also allow for nonlinear effects and find some evidence that employment losses accelerate with larger NTR gaps, rather than reflecting a fragile linear specification.

4.4 Instrument-based robustness

- The paper estimates a two-stage least squares robustness check that instruments the DID term $\text{PostPNTR}_t \times \text{NTR Gap}_i$ with $\text{PostPNTR}_t \times \text{Non-NTR Rate}_i$.
- **Logic.** This leans even more heavily on the historically predetermined Smoot-Hawley tariff schedule.
- **Takeaway.** The IV estimate remains negative and statistically significant, and if anything is somewhat larger in magnitude than the baseline OLS estimate.

4.5 The European comparison

- The EU comparison is a particularly strong check because it confronts the main story with a clear counterfactual.
- China did *not* experience the same policy change vis-à-vis the EU in the early 2000s, so a similar employment pattern there would weaken the PNTR interpretation.
- Instead, the paper finds the opposite: the negative gap-employment relationship is present in the United States and absent in the European Union.

4.6 Relation to the literature

- The paper is closely related to the “China shock” literature, but its mechanism is distinct. It emphasizes the resolution of **trade policy uncertainty**, not only realized import penetration.
- It also connects to the literature on investment under uncertainty, showing that firms’ responses to the removal of downside policy risk can be large and fast.
- More broadly, it helps explain why manufacturing employment can drop sharply even when manufacturing value added does not collapse by the same amount.

5 Tables and figures

5.1 Figure 1: China MFN/NTR uncertainty index

Read:

- The figure shows that uncertainty about China’s trade status with the United States was real, salient, and time-varying before PNTR.
- The index spikes in periods of political tension, especially in the early 1990s and around the 2000 PNTR debate.
- After PNTR passes, the index essentially falls to zero.
- This figure motivates the entire paper: the relevant policy shock is the elimination of uncertainty, not a contemporaneous cut in applied tariff rates.

5.2 Figure 2: Distribution of NTR gaps across industries

Read:

- There is substantial cross-industry variation in exposure to PNTR.
- In 1999, the average NTR gap is roughly **33 percentage points**, with a standard deviation of about **14 percentage points**.
- This dispersion is what identifies the DID design: some industries had much more uncertainty removed than others.
- The figure also makes clear that the treatment is continuous rather than binary.

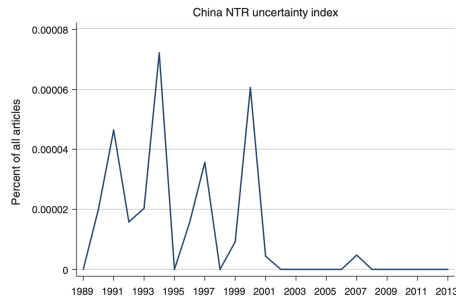


FIGURE 1. CHINA MOST FAVORED NATION (MFN) UNCERTAINTY INDEX

Note: Figure displays percent of *New York Times*, *Wall Street Journal*, and *Washington Post* articles discussing the uncertainty of China's NTR status.

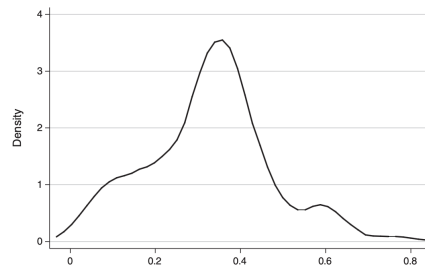


FIGURE 2. DISTRIBUTION OF NTR GAPS ACROSS CONSTANT MANUFACTURING INDUSTRIES, 1999

5.3 Figure 3: Manufacturing employment versus value added

Read:

- US manufacturing employment drops sharply after 2000, but real manufacturing value added does not collapse in parallel.
- This divergence suggests that the key margin is not just less manufacturing output, but also **less labor used per unit of manufacturing activity**.
- That observation helps motivate the later analysis of capital deepening, skill intensity, and plant-level adjustment.
- The paper is therefore not only about import competition displacing output; it is also about import competition reshaping how remaining US production is carried out.

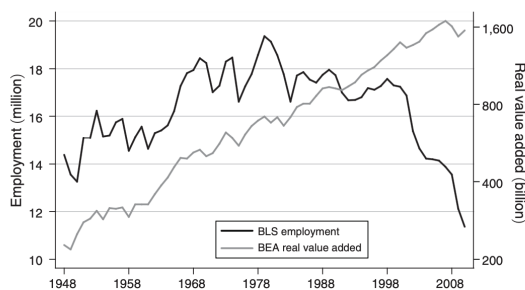


FIGURE 3. US MANUFACTURING EMPLOYMENT VERSUS VALUE ADDED

Note: Figure compares annual manufacturing employment as of March according to the US Bureau of Labor Statistics (1948–2010, series CEU3000000001) to real value added as measured by the Bureau of Economic Analysis (1948–2010).

TABLE 1—BASELINE RESULTS (*LBD*)

	$\ln(\text{Emp}_{it})$	$\ln(\text{Emp}_{it})$	$\ln(\text{Emp}_{it})$
Post $\times$ NTR Gap _{<i>i</i>}	−0.714 (0.193)	−0.601 (0.191)	−0.469 (0.147)
Post $\times \ln(K/\text{Emp}_{i,1990})$		0.037 (0.031)	−0.016 (0.025)
Post $\times \ln(NP/\text{Emp}_{i,1990})$		0.081 (0.054)	0.132 (0.053)
Post $\times$ Contract Intensity _{<i>i</i>}			−0.181 (0.112)
Post $\times \Delta$ China Import Tariffs _{<i>i</i>}			−0.244 (0.140)
Post $\times \Delta$ China Subsidies _{<i>i</i>}			0.063 (0.088)
Post $\times \Delta$ China Licensing _{<i>i</i>}			−0.238 (0.164)
Post $\times 1\{\text{Advanced Technology}_i\}$			−0.036 (0.045)
MFA Exposure _{<i>it</i>}			−0.342 (0.060)
NTR _{<i>it</i>}			−0.455 (0.670)
US Union Membership _{<i>it</i>}			−0.123 (0.203)
Observations	5,700	5,700	5,700
R ²	0.98	0.98	0.99
Fixed effects	<i>it</i>	<i>it</i>	<i>it</i>
Employment weighted	Yes	Yes	Yes
Implied impact of PNTR	−0.229	−0.193	−0.151

Note: Table reports results of OLS generalized difference-in-differences regressions. The dependent variable is the log of industry-year employment and the independent variable representing the effect of PNTR is the interaction of the NTR gap and a post-PNTR indicator. Additional controls include time-varying variables—MFA exposure, NTR tariff rates, union membership rates—as well as interactions of the post-PNTR indicator with time-invariant controls including the log of 1990 capital and skill intensity, contract intensity (Nunn 2007), changes in Chinese import tariffs, changes in Chinese production subsidies, changes in Chinese export licensing requirements, and an indicator for whether the industry produces advanced technology products. Data span 1990 to 2007. Robust standard errors adjusted for clustering at the industry (*i*) level are displayed below each coefficient. Estimates for the year (*t*) and industry fixed effects as well as the constant are suppressed. Observations are weighted by 1990 industry employment. Final row reports the predicted relative change in the dependent variable implied by the difference-in-differences coefficient. Number of observations has been rounded to nearest thousand due to Census Bureau disclosure avoidance procedures.

5.4 Table 1: Baseline employment results

Read:

- The DID coefficient on $\text{PostPNTR} \times \text{NTR Gap}$ is negative and statistically significant in every specification.
- In the baseline column, the coefficient is about -0.469 with a standard error of **0.147**.
- Moving an industry from the 25th to the 75th percentile of the NTR-gap distribution implies an additional relative employment loss of roughly **0.08 log points**.
- The employment-weighted implied average effect of PNTR in the baseline is about -0.151 **log points**, so the result is economically large, not just statistically significant.

5.5 Figure 4 and Table 2: Timing and robustness

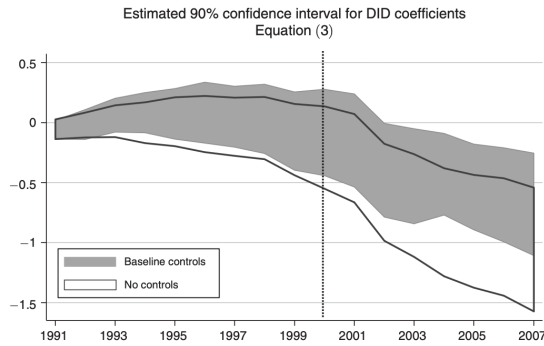


FIGURE 4. ESTIMATED TIMING OF THE PNTR EFFECT (LBD)

Notes: Figure displays the 90 percent confidence level interval (CI) for the estimated difference-in-differences (DID) coefficients for interactions of year dummies with the NTR gap from equation (3). Shaded CI represents the specification which includes all baseline covariates. Unshaded CI represents the specification which includes only the DID coefficients and the fixed effects. Baseline covariates include time-varying variables—MFA exposure, NTR tariff rates, union membership rates—as well as interactions of year dummies with time invariant controls including the log of 1990 capital and skill intensity, contract intensity (Nunn 2007), changes in Chinese import tariffs, changes in Chinese production subsidies, changes in Chinese export licensing requirements, and an indicator for whether the industry produces advanced technology products. Observations are weighted by 1990 industry employment. Confidence interval is based on robust standard errors adjusted for clustering at the industry level.

Read:

- The event-study figure shows no meaningful pre-trend before 2001, which is central for the DID interpretation.

TABLE 2—ROBUSTNESS EXERCISES (LBD)

	$\ln(\text{Emp}_{it})$	$\ln(\text{Emp}_{it})$	$\ln(\text{Emp}_{it})$	$\ln(\text{Emp}_{it})$	$\ln(\text{Emp}_{it})$	$\ln(\text{Emp}_{it})$
Post $\times$ NTR Gap _i	-0.617 (0.152)		-0.244 (0.429)		-0.475 (0.147)	-0.461 (0.146)
Post $\times$ 1990 NTR Gap _i		-0.410 (0.166)				-0.484 (0.164)
Post $\times$ NTR Gap ₁₂			-0.346 (0.636)			
Post $\times$ NTR Gap _i (Slope 1)				-0.460 (0.513)		
Post $\times$ NTR Gap _i (Intercept 2)				0.251 (0.630)		
Post $\times$ NTR Gap _i (Slope 2)				-1.683 (1.054)		
Post $\times$ $\ln(K/\text{Emp}_{i,1990})$	-0.026 (0.023)	-0.020 (0.026)	-0.015 (0.025)	0.414 (0.204)	-0.044 (0.018)	-0.009 (0.027)
Post $\times$ $\ln(NP/\text{Emp}_{i,1990})$	0.129 (0.049)	0.132 (0.050)	0.130 (0.053)	-0.511 (0.283)	0.186 (0.047)	0.130 (0.052)
Post $\times$ Contract Intensity _i	-0.191 (0.103)	-0.154 (0.108)	-0.184 (0.114)	1.306 (0.738)	-0.178 (0.112)	-0.058 (0.120)
Post $\times$ Δ China Import Tariffs _i	-0.230 (0.136)	-0.282 (0.142)	-0.269 (0.144)	-0.557 (0.489)	-0.248 (0.140)	-0.248 (0.140)
Post $\times$ Δ China Subsidies _i	0.068 (0.085)	0.051 (0.086)	0.073 (0.092)	-1.650 (0.691)	0.064 (0.088)	0.066 (0.088)
Post $\times$ Δ China Licensing _i	-0.220 (0.155)	-0.217 (0.161)	-0.237 (0.166)	-1.319 (0.658)	-0.240 (0.164)	-0.236 (0.165)
Post $\times$ $I(\text{Advanced Technology})_i$	-0.033 (0.044)	-0.050 (0.045)	-0.036 (0.045)	0.018 (0.046)	-0.038 (0.045)	-0.036 (0.045)
MFA Exposure _{it}	-0.337 (0.057)	-0.347 (0.058)	-0.342 (0.061)	-0.541 (0.198)	-0.342 (0.063)	-0.345 (0.059)
NTR _{it}	-0.578 (0.656)	-0.501 (0.852)	-0.430 (0.672)	24.346 (13.226)	-0.724 (0.672)	-0.448 (0.670)
US Union Membership _{it}	-0.124 (0.195)	-0.123 (0.203)	-0.110 (0.199)	15.049 (7.792)	-0.086 (0.200)	-0.124 (0.203)
$\ln(\text{RGDP}_{it}) \times \ln(K/\text{Emp}_{i,1990})$					0.104 (0.097)	
$\ln(\text{RGDP}_{it}) \times \ln(NP/\text{Emp}_{i,1990})$					-0.193 (0.181)	
Treffer Business Cycle _{it}					0.242 (0.106)	
Revealed NTR _{it}						0.495 (0.291)
Observations	5,700	5,700	5,700	5,700	5,700	5,700
R ²	0.99	0.99	0.99	0.99	0.99	0.99
Estimation	2SLS	OLS	OLS	OLS	OLS	OLS
Instrument	NTR rate					
Fixed effects	i	i	i	i	i	i
Employment weighted	Yes	Yes	Yes	Yes	Yes	Yes
Joint significance p-value				0.008	0.011	
Constrained estimation?	No	No	No	Yes	No	No
Knot				0.45		
AIC				13,810		
Implied impact of PNTR	-0.198	-0.132	-0.120	-0.158	-0.153	-0.148

Notes: Table reports results of OLS generalized difference-in-differences regressions of log industry employment on noted permutations of an interaction of the NTR gap with an indicator for the post-PNTR period. Additional controls include time-varying variables—MFA exposure, NTR tariff rates, union membership rates—as well as interactions of the post-PNTR indicator with time-invariant controls including the log of 1990 capital and skill intensity, contract intensity (Nunn 2007), changes in Chinese import tariffs, changes in Chinese production subsidies, changes in Chinese export licensing requirements, and an indicator for whether the industry produces advanced technology products. Data span 1990 to 2007. Robust standard errors adjusted for clustering at the industry (i) level are displayed below each coefficient. Estimates for the year (t) and industry fixed effects as well as the constant are suppressed. Observations are weighted by 1990 industry employment. “Joint significance” reports the p -value of a test of joint significance of the linear and polynomial NTR gap terms in column 3 and the intercept and slope terms in column 4. Penultimate row reports the Akaike information criterion (AIC) for the two-segment spline estimated in the fourth column. Final row reports the predicted relative change in the dependent variable implied by the difference-in-differences coefficient. Number of observations has been rounded to nearest thousand due to Census Bureau disclosure avoidance procedures.

- After 2001, the coefficients turn negative and grow more negative over time.
- Table 2 shows that the result survives IV estimation, use of the 1990 NTR gap, nonlinear specifications, business-cycle controls, and use of revealed tariffs.
- The message is that the baseline finding is not an artifact of one narrow specification choice.

5.6 Table 3: United States versus European Union

Read:

- The triple-difference coefficient comparing the United States to the EU is negative and statistically significant, around -0.641 .
- Separate DID regressions show a strong negative effect in the United States but essentially no effect in the European Union.
- This is one of the cleanest pieces of evidence in the paper because the EU did not experience the same policy change.
- The table therefore cuts against explanations based on global manufacturing shocks or a generic export surge from China affecting all destinations the same way.

	$\ln(\text{Emp}_u)$	$\ln(\text{Emp}_e)$	$\ln(\text{Emp}_a)$
Post $\times$ NTR Gap ₁₉₉₉ $\times$ 1{c = US}	-0.641 (-0.247)		
Post $\times$ NTR Gap ₁₉₉₉		0.016 (-0.112)	-0.649 (-0.270)
Observations	1,664	999	832
R ²	0.997	0.994	0.982
Fixed effects	ct, ct, it	it	it
Employment weighted	Yes	Yes	Yes

Notes: First column displays results of an OLS generalized triple differences regression of the log of employment on a triple interaction of an indicator for the post-PNTR period, the NTR gap and an indicator for if the country is the United States. The countries included are the United States and European Union. The second and third columns display results of industry-level generalized difference-in-differences regressions for the European Union and United States, respectively. Data span 1998 to 2005. Estimates for country $\times$ year (ct), country $\times$ industry (ct), and industry $\times$ year (it) fixed effects (for column 1), and industry and year fixed effects (for columns 2 and 3), as well as the regression constant are not reported. Robust standard errors adjusted for clustering at the country $\times$ industry-level (for column 1) and industry-level (for columns 2 and 3) are displayed below each coefficient. Observations are weighted by 1998 employment.

	$\ln(\text{Value}_{hct})$	$\ln(\text{Importers}_{hct})$	$\ln(\text{Exporters}_{hct})$	$\ln(\text{Pairs}_{hct})$
1{c = China} $\times$ Post $\times$ NTR Gaph	0.476 (0.102)	0.461 (0.051)	0.515 (0.058)	0.517 (0.061)
$\ln(\text{Tariff}_{hct})$	0.039 (0.010)	0.024 (0.003)	0.026 (0.003)	0.028 (0.004)
MFA phase 1 _{hct}	0.100 (0.058)	0.039 (0.021)	0.067 (0.024)	0.072 (0.026)
MFA phase 2 _{hct}	-0.101 (0.036)	-0.021 (0.013)	-0.016 (0.015)	-0.030 (0.016)
MFA phase 3 _{hct}	-0.066 (0.039)	0.070 (0.014)	0.053 (0.016)	0.041 (0.017)
MFA phase 4 _{hct}	-0.452 (0.026)	0.034 (0.009)	-0.069 (0.011)	-0.085 (0.011)
Observations	1,396,000	1,396,000	1,396,000	1,396,000
R ²	0.84	0.91	0.91	0.91
Fixed effects	hc, ct, ht	hc, ct, ht	hc, ct, ht	hc, ct, ht
Implied impact of PNTR	0.17	0.17	0.19	0.19

Notes: Table displays results of product (h)-country (c)-year (t) level OLS generalized triple difference-in-differences regression of noted dependent variable on interaction of China country indicator, post-PNTR indicator, and NTR gap, along with country $\times$ year (ct), country $\times$ product (hc) and product $\times$ year (ht) fixed effects, the revealed tariff rate and indicators for the four phases of the MFA. Data span 1992 to 2007. Robust standard errors adjusted for clustering at the product $\times$ country level are displayed below each coefficient. Estimates for the fixed effects and constant are suppressed. Final row reports the predicted relative change in the dependent variable implied by the triple differences coefficient. Number of observations has been rounded to nearest thousand due to Census Bureau disclosure avoidance procedures.

5.7 Table 4: PNTR and US imports from China

Read:

- Products with larger NTR gaps experience larger post-PNTR increases in US imports from China relative to imports from other countries.
- The response is broad-based: import value rises, and so do the number of US importers, the number of Chinese exporters, and the number of importer-exporter pairs.
- The implied effects are on the order of roughly **0.17 to 0.19 log points** across these margins.
- This table shows that the employment result is matched by the expected trade response: high-gap industries face a larger import shock from China after uncertainty is removed.

5.8 Table 5: PNTR and Chinese exports

Read:

- Chinese exports to the United States rise more in products with larger NTR gaps, relative to exports to other destinations.
- The response is strongest among **foreign-owned Chinese firms**, where the coefficient in all-trade exports is about **1.018**, much larger than for SOEs.
- The same pattern appears in general trade and in processing-and-assembly trade.
- This is consistent with offshoring or multinational production responses, not only expansion by purely domestic Chinese firms.

TABLE 5—PNTR AND CHINESE EXPORTS (*Chinese Data*)

	All firms $\ln(V_{hct})$	SOE $\ln(V_{hct})$	Domestic $\ln(V_{hct})$	Foreign $\ln(V_{hct})$
<i>Panel A. All trade</i>				
$1\{c = \text{US}\} \times \text{Post} \times \text{NTR Gap}_h$	0.214 (0.126)	0.187 (0.138)	0.815 (0.473)	1.018 (0.207)
Observations	1,159,132	972,780	473,590	510,839
R^2	0.84	0.75	0.71	0.76
Fixed effects	hc, ht, ct	hc, ht, ct	hc, ht, ct	hc, ht, ct
<i>Panel B. General trade</i>				
$1\{c = \text{US}\} \times \text{Post} \times \text{NTR Gap}_h$	0.357 (0.129)	0.333 (0.150)	0.728 (0.475)	0.919 (0.226)
Observations	1,112,173	945,902	467,854	419,451
R^2	0.82	0.81	0.81	0.79
Fixed effects	hc, ht, ct	hc, ht, ct	hc, ht, ct	hc, ht, ct
<i>Panel C. Processing and assembly trade</i>				
$1\{c = \text{US}\} \times \text{Post} \times \text{NTR Gap}_h$	0.287 (0.176)	0.185 (0.211)	1.802 (2.086)	0.707 (0.209)
Observations	344,604	182,274	39,444	275,940
R^2	0.86	0.85	0.85	0.85
Fixed effects	hc, ht, ct	hc, ht, ct	hc, ht, ct	hc, ht, ct

Notes: Table displays results of product (h)-country (c)-year (t) level OLS generalized triple difference-in-differences regression of log Chinese export value on interaction of US country indicator, post-PNTR indicator, and NTR gap, along with country $\times$ year (ct), country $\times$ product (hc), and product $\times$ year (ht) fixed effects, and the NTR tariff rate. Data span 2000 to 2005. Robust standard errors adjusted for clustering at the product $\times$ country level are displayed below each coefficient. Estimates for the fixed effects and constant are suppressed.

TABLE 6—PNTR AND RELATED-PARTY US IMPORTS (*LFTTD*)

	$\ln(\text{RP Value}_{hct})$	$\ln(\text{RP Importers}_{hct})$	$\ln(\text{RP Exporters}_{hct})$	$\ln(\text{RP Pairs}_{hct})$
$1\{c = \text{China}\} \times \text{Post}$	0.205 (0.180)	0.328 (0.063)	0.379 (0.074)	0.380 (0.074)
$\times \text{NTR Graph}$				
$\ln(\text{Tariff}_{hct})$	-0.045 (0.016)	0.001 (0.004)	0.002 (0.005)	0.002 (0.005)
MFA Phase 1 $_{hct}$	-0.151 (0.103)	0.005 (0.030)	0.022 (0.035)	0.016 (0.036)
MFA Phase 2 $_{hct}$	-0.246 (0.067)	-0.049 (0.019)	-0.030 (0.022)	-0.046 (0.022)
MFA Phase 3 $_{hct}$	-0.109 (0.073)	0.004 (0.020)	0.033 (0.025)	0.021 (0.025)
MFA Phase 4 $_{hct}$	-0.617 (0.049)	0.029 (0.014)	-0.150 (0.017)	-0.154 (0.017)
Observations	712,000	712,000	712,000	712,000
R^2	0.82	0.88	0.87	0.87
Fixed effects	hc, ct, ht	hc, ct, ht	hc, ct, ht	hc, ct, ht
Implied impact of PNTR	0.06	0.10	0.12	0.12

Notes: Table displays results of product (h)-country (c)-year (t) level OLS generalized triple difference-in-differences regression of related-party trade variable on interaction of China country indicator, post-PNTR indicator, and NTR gap, along with country $\times$ year (ct), country $\times$ product (hc) and product $\times$ year (ht) fixed effects, the revealed tariff rate and indicators for the four phases of the MFA. Data span 1992 to 2007. Robust standard errors adjusted for clustering at the product $\times$ country level are displayed below each coefficient. Estimates for the fixed effects and constant are suppressed. Final row reports the predicted relative change in the dependent variable implied by the triple differences coefficient. Number of observations has been rounded to nearest thousand due to Census Bureau disclosure avoidance procedures.

5.9 Table 6: Related-party US imports

Read:

- Related-party trade relationships with China expand after PNTR in the products with the largest NTR gaps.
- The number of related-party US importers, Chinese exporters, and importer-exporter pairs all rise significantly; the implied impacts are roughly **0.10 to 0.12 log points**.
- The value margin is positive but less precisely estimated, which the authors interpret as reflecting lags between forming relationships and moving substantial trade volumes through them.
- The table provides direct evidence consistent with **offshoring within firm boundaries**.

5.10 Tables 7 and 8: Factor intensity and continuing plants

Read:

- At the industry level, PNTR is associated with higher **skill intensity** and higher **capital intensity**.

	ln(NP/Emp _{it})	ln(K/Emp _{it})	ln(NProd _{it})	ln(Prod _{it})	ln(Emp _{it})	ln(K _{it})	ln(Hours _{it})
Post × NTR Gap _{it}	0.159 (0.062)	0.556 (0.157)	-0.324 (0.186)	-0.531 (0.222)	-0.481 (0.218)	-0.097 (0.230)	-0.567 (0.228)
Post × ln(K/Emp _{it,1990})	-0.049 (0.018)	-0.031 (0.031)	0.019 (0.042)	0.006 (0.039)	-0.082 (0.041)	0.010 (0.041)	-0.017 (0.041)
Post × ln(NP/Emp _{it,1990})		-0.021 (0.054)	0.075 (0.077)	0.217 (0.079)	0.181 (0.079)	0.200 (0.083)	0.223 (0.081)
Post × Contract Intensity _{it}	0.134 (0.045)	0.172 (0.130)	-0.017 (0.159)	-0.299 (0.178)	-0.201 (0.169)	-0.193 (0.158)	-0.322 (0.179)
Post × ΔChina Import Tariffs _{it}	0.127 (0.066)	-0.213 (0.159)	-0.142 (0.206)	-0.467 (0.217)	-0.332 (0.213)	-0.507 (0.212)	-0.495 (0.228)
Post × ΔChina Subsidies _{it}	0.061 (0.054)	0.298 (0.085)	0.140 (0.106)	0.015 (0.133)	0.070 (0.121)	0.401 (0.129)	-0.017 (0.143)
Post × ΔChina Licensing _{it}	-0.051 (0.071)	0.065 (0.138)	-0.314 (0.232)	-0.222 (0.250)	-0.265 (0.235)	-0.095 (0.229)	-0.156 (0.262)
Post × I{Advanced Technology _{it} }	-0.044 (0.019)	-0.015 (0.055)	-0.010 (0.068)	-0.040 (0.075)	-0.018 (0.070)	-0.021 (0.070)	-0.043 (0.074)
MFA Exposure _{it}	0.072 (0.060)	0.048 (0.117)	-0.274 (0.080)	-0.352 (0.132)	-0.335 (0.117)	-0.306 (0.103)	-0.344 (0.132)
NTR _{it}	-0.108 (0.494)	0.182 (0.959)	-1.246 (1.114)	-0.696 (1.406)	-1.094 (1.280)	-0.709 (1.147)	-0.781 (1.417)
US Union Membership _{it}	0.298 (0.148)	0.241 (0.263)	0.170 (0.376)	-0.292 (0.347)	-0.158 (0.359)	-0.113 (0.354)	-0.273 (0.362)
Observations	1,280	1,280	1,280	1,280	1,280	1,280	1,280
R ²	0.97	0.97	0.98	0.98	0.98	0.98	0.98
Fixed effects	<i>it</i>	<i>it</i>	<i>it</i>	<i>it</i>	<i>it</i>	<i>it</i>	<i>it</i>
Employment weighted	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Implied impact of PNTR	0.051	0.178	-0.104	-0.170	-0.154	-0.031	-0.182

Notes: Table reports results of industry-year level OLS generalized difference-in-differences regression of noted industry outcome on the interaction of the NTR gap and a post-PNTR indicator. Additional controls include time-varying variables—MFA exposure, NTR tariff rates, union membership rates—as well as interactions of the post-PNTR indicator with time-invariant controls including the log of 1990 capital and skill intensity, contract intensity (Nunn 2007), changes in Chinese import tariffs, changes in Chinese production subsidies, changes in Chinese export licensing requirements, and an indicator for whether the industry produces advanced technology products. NProd, Prod, Hours, and K represent nonproduction workers, production workers, production hours, and the real book value of capital. Data are from the CM and for census years 1992, 1997, 2002, and 2007. Robust standard errors adjusted for clustering at the industry level are displayed below each coefficient. Estimates for the year (*t*) and industry (*i*) fixed effects as well as the constant are suppressed. Observations are weighted by 1992 industry employment. Final row reports the predicted relative change in the dependent variable implied by the difference-in-differences coefficient.

	ln(NP/Emp _{it})	ln(K/Emp _{it})	ln(NProd _{it})	ln(Prod _{it})	ln(Emp _{it})	ln(K _{it})	ln(Hours _{it})
Post × NTR Gap _{it}	0.003 (0.082)	0.295 (0.139)	-0.240 (0.123)	-0.280 (0.142)	-0.276 (0.112)	-0.419 (0.268)	-0.312 (0.139)
Post × ln(K/Emp _{it,1990})	-0.039 (0.006)		-0.027 (0.013)	-0.008 (0.014)	-0.009 (0.012)	-0.254 (0.023)	-0.017 (0.013)
Post × ln(NP/Emp _{it,1990})		0.013 (0.021)	-0.290 (0.024)	0.109 (0.020)	0.016 (0.018)	0.070 (0.028)	0.106 (0.020)
Age _{it}	0.022 (0.017)	0.092 (0.026)	0.197 (0.031)	0.213 (0.030)	0.195 (0.025)	0.244 (0.027)	0.216 (0.027)
TFP _{it}	-0.042 (0.012)	-0.073 (0.021)	-0.007 (0.012)	0.032 (0.022)	0.031 (0.017)	-0.037 (0.025)	0.046 (0.023)
Post × Contract Intensity _{it}	0.034 (0.058)	0.114 (0.097)	0.064 (0.117)	-0.214 (0.128)	-0.094 (0.102)	-0.369 (0.168)	-0.251 (0.123)
Post × ΔChina Import Tariffs _{it}	0.236 (0.079)	-0.166 (0.168)	0.222 (0.165)	-0.388 (0.190)	-0.229 (0.144)	-0.294 (0.288)	-0.436 (0.164)
Post × ΔChina Subsidies _{it}	0.014 (0.052)	0.239 (0.103)	0.150 (0.097)	0.141 (0.123)	0.103 (0.086)	0.453 (0.174)	0.103 (0.111)
Post × ΔChina Licensing _{it}	0.180 (0.072)	0.155 (0.170)	-0.210 (0.201)	-0.550 (0.252)	-0.387 (0.188)	0.064 (0.267)	-0.460 (0.214)
Post × I{Advanced Technology _{it} }	-0.048 (0.020)	-0.015 (0.054)	0.051 (0.043)	-0.130 (0.063)	-0.057 (0.048)	-0.013 (0.079)	-0.107 (0.063)
MFA Exposure _{it}	0.085 (0.058)	0.081 (0.066)	-0.139 (0.089)	-0.191 (0.100)	-0.185 (0.095)	-0.184 (0.116)	-0.201 (0.101)
NTR _{it}	-0.849 (0.582)	-0.949 (0.681)	0.015 (0.844)	1.827 (1.220)	0.868 (0.903)	0.904 (1.080)	1.666 (1.208)
US Union Membership _{it}	0.268 (0.145)	-0.009 (0.240)	0.591 (0.237)	0.179 (0.252)	0.304 (0.223)	-0.262 (0.263)	0.109 (0.259)
Observations	257,503	257,503	257,503	257,503	257,503	257,503	257,503
R ²	0.78	0.73	0.93	0.93	0.95	0.91	0.92
Fixed effects	<i>p,t</i>	<i>p,t</i>	<i>p,t</i>	<i>p,t</i>	<i>p,t</i>	<i>p,t</i>	<i>p,t</i>
Employment weighted	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Implied impact of PNTR	0.001	0.094	-0.077	-0.089	-0.088	-0.134	-0.100

Notes: Table reports results of plant-year level OLS generalized difference-in-differences regression of noted plant outcome on the interaction of the NTR gap and a post-PNTR indicator. Additional controls include time-varying variables—MFA exposure, NTR tariff rates, union membership rates, plant age, plant TFP (index method)—as well as interactions of the post-PNTR indicator with time-invariant controls including the log of capital and skill intensity in the first year the plant is observed, contract intensity (Nunn 2007), changes in Chinese import tariffs, changes in Chinese production subsidies, changes in Chinese export licensing requirements, and an indicator for whether the industry produces advanced technology products. NProd, Prod, Hours, and K represent non-production workers, production workers, production hours and the real book value of capital. Data are from the CM and for census years 1992, 1997, 2002, and 2007. Robust standard errors adjusted for clustering at the level of the plants' major industries are displayed below each coefficient. Estimates for the year (*t*) and plant (*p*) fixed effects as well as the constant are suppressed. Observations are weighted by employment in the first year the plant is observed. Final row reports the predicted relative change in the dependent variable implied by the difference-in-differences coefficient.

- Employment falls for both non-production and production workers, but the decline is larger for **production workers**, which fits the idea that routine production jobs are most exposed.
- Within continuing plants, there is little evidence of skill deepening, but there is evidence of **capital deepening**.
- So the overall industry response reflects both reallocation across plants and technological or product-mix adjustment within surviving plants.

5.11 Table 9: Input-output linkages and plant death

Read:

- Plants directly exposed to larger NTR gaps shrink more and are more likely to exit after PNTR.
- Plants whose **downstream customers** are more exposed also contract and are more likely to die.
- By contrast, upstream exposure is weak, while downstream exposure is economically important.
- This means the employment effects of PNTR propagate through supply chains rather than staying confined to directly treated industries.

TABLE 9—PLANT INPUT-OUTPUT LINKAGES (*LBD*)

	$\ln(\text{Emp}_{pt})$	$\ln(\text{Emp}_p)$	$1\{\text{Death}_{pt+1}\}$	$1\{\text{Death}_{pt+1}\}$
Post $\times$ NTR Gap _{<i>p</i>}	-0.380 (0.089)	-0.208 (0.090)	0.064 (0.020)	0.042 (0.019)
Post $\times$ NTR Gap _{<i>p</i>} ^{Upstream}		-0.280 (0.427)		-0.022 (0.082)
Post $\times$ NTR Gap _{<i>p</i>} ^{Downstream}		-0.691 (0.159)		0.103 (0.041)
Post $\times \ln(K/\text{Emp}_{p,1990})$	-0.082 (0.016)	-0.070 (0.015)	-0.006 (0.003)	-0.009 (0.003)
Post $\times \ln(NP/\text{Emp}_{p,1990})$	0.052 (0.031)	0.034 (0.030)	-0.016 (0.006)	-0.013 (0.005)
Post $\times$ Contract Intensity _{<i>i</i>}	-0.189 (0.081)	-0.218 (0.081)	0.003 (0.013)	0.010 (0.014)
Post $\times \Delta\text{China Import Tariffs}_i$	-0.396 (0.104)	-0.278 (0.109)	-0.006 (0.020)	-0.027 (0.018)
Post $\times \Delta\text{China Subsidies}_i$	0.022 (0.073)	0.022 (0.069)	0.023 (0.012)	0.023 (0.011)
Post $\times \Delta\text{China Licensing}_i$	-0.121 (0.146)	-0.036 (0.135)	0.007 (0.022)	-0.002 (0.023)
Post $\times 1\{\text{Advanced Technology}_i\}$	-0.056 (0.028)	-0.055 (0.027)	0.007 (0.005)	0.005 (0.004)
MFA Exposure _{<i>i</i>}	-0.193 (0.064)	-0.167 (0.053)	0.037 (0.014)	0.035 (0.013)
NTR _{<i>it</i>}	0.555 (0.513)	0.524 (0.513)	0.031 (0.052)	0.039 (0.048)
US Union Membership _{<i>it</i>}	0.112 (0.132)	0.164 (0.133)	0.012 (0.014)	0.003 (0.012)
Observations	1,181,142	1,181,142	2,079,616	2,079,616
R^2	0.95	0.95	0.79	0.79
Fixed effects	<i>p,t</i>	<i>p,t</i>	<i>p,t</i>	<i>p,t</i>
Employment weighted	Yes	Yes	Yes	Yes
Implied impact of PNTR	-0.117	-0.143	—	—

Notes: Table reports results of plant-year level OLS generalized difference-in-differences regressions of either log plant-year employment or an indicator for plant death on the interaction of a post-PNTR indicator and the own, upstream and downstream NTR gaps. Additional controls include time-varying variables—MFA exposure, NTR tariff rates, union membership rates, plant age, plant TFP (index method)—as well as interactions of the post-PNTR indicator with time-invariant controls including the log of capital and skill intensity in the first year the plant is observed, contract intensity (Nunn 2007), changes in Chinese import tariffs, changes in Chinese production subsidies, changes in Chinese export licensing requirements and an indicator for whether the industry produces advanced technology products. Data span 1990 to 2007. Robust standard errors adjusted for clustering at the level of the plants' major industries are displayed below each coefficient. Estimates for the year (*t*) and plant (*p*) fixed effects as well as the constant are suppressed. Observations are weighted by plant employment in the first year that it is observed in the sample. Final row reports the predicted relative change in the dependent variable implied by the difference-in-differences coefficient. First two columns restricted to the intensive margin of plants active in all years of the sample. Last two columns contain all observations.

6 Short summary

- **Core finding.** The sharp fall in US manufacturing employment after 2000 is linked to the United States' grant of PNTR to China.
- **Mechanism.** PNTR removed the possibility of large future tariff increases on Chinese imports, encouraging import growth, Chinese market entry, offshoring, and less labor-intensive domestic production.
- **Identification.** The design compares industries with different historical NTR gaps before and after PNTR, checks for pre-trends, controls for many alternative explanations, and uses the EU as an external comparison group.
- **Broader lesson.** Trade policy uncertainty can have large real effects even when current tariff rates do not move.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a robust empirical connection between China's receipt of PNTR and the collapse of US manufacturing employment in the 2000s.
- The key channel is not a contemporaneous tariff cut, but the removal of uncertainty about future tariff treatment.

- Industries more exposed to this policy change lose more employment and see stronger trade responses from China.

7.2 Economic intuition

- Before PNTR, firms contemplating China-related investments faced a downside risk: if annual renewal failed, tariffs could jump from NTR to much higher non-NTR levels.
- When PNTR removes that risk, projects that previously looked too risky become profitable: import relationships are formed, Chinese exporters enter, US firms shift sourcing and production, and domestic plants economize on labor.
- Because these are sunk-cost decisions, the response can be fast and discrete once uncertainty is resolved.

7.3 Takeaway

This paper demonstrates that *uncertainty about trade policy* can matter as much as realized tariff changes. Its main contribution is to show that the early-2000s decline in US manufacturing employment is not just a story about more imports from China in a generic sense. It is a story about a specific institutional change that permanently altered firms' expectations, triggered new investment and sourcing decisions, and then spread through production networks and plant-level adjustment. For work on the China shock, policy uncertainty, offshoring, and the reorganization of production, this paper is a benchmark.